

MOKD-EEG: Multi-task Learning-based Online Knowledge Distillation for Cross-task EEG Classification

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Abstract—Electroencephalogram (EEG)-based Multi-task learning has shown great potential in Brain-Computer Interface (BCI) research. However, Cross-task classification remains challenging due to the diverse and task-specific characteristics of EEG signals in Motor Imagery and Emotion Recognition tasks. To address this challenge, this paper propose a Multi-task Learning-based Online Knowledge Distillation (MOKD-EEG) framework for Cross-task to enhance the adaptability of EEG classification models. Initially, EEG data is first preprocessed using Short-Time Fourier Transform (STFT) to ensure compatibility with feature extraction networks. In addition, the framework introduces Online Knowledge Distillation Training (OKD-T) to optimize task-specific and shared representations, which incorporates Single-task and Multi-task networks. To further mitigate task conflicts, Adaptive Representation Distillation (ARD) is employed to dynamically align intermediate features across tasks. Finally, Dynamic Weight Adjustment (DWA) is introduced to prioritize tasks based on their learning states, ensuring balanced optimization during training. Experimental evaluations on multiple public datasets demonstrate that MOKD-EEG significantly improves robustness compared to existing methods.

Index Terms—Electroencephalogram (EEG), Cross-task learning, Multi-task learning (MTL), Brain-Computer Interface (BCI)

I. INTRODUCTION

Electroencephalogram (EEG) classification is a foundational task in brain-computer interface (BCI) systems, facilitating direct interaction between the brain and external devices. EEG has applications in diverse areas such as medical diagnostics, motor imagery, and cognitive state monitoring, offering significant opportunities for individuals with motor disabilities and advancing neurological research [1]–[3]. Recent research highlights that EEG signals can be used effectively in classifying mental states across various tasks, showcasing the potential of BCIs in adapting to diverse applications [4]. However, the high-dimensionality and non-stationary characteristics of EEG signals with substantial variability in different tasks make cross-task EEG classification a particularly challenging problem [5]. This challenge highlights the need for robust models

capable of generalization across diverse datasets, paving the way for BCIs widely applicable in real-world applications [6].

Multi-task learning is a promising method for improving EEG classification. It helps overcome limitations in traditional approaches. This method uses shared representations to enable models to generalize effectively across diverse tasks. Multi-task learning has shown success in domains like computer vision and natural language processing, where shared features between tasks improve overall performance [7]. In EEG-based BCIs, recent advancements have demonstrated the effectiveness of cross-task frameworks in improving adaptability by employing domain-invariant feature extraction and adaptation techniques [8]. Negative transfer is also a well-documented challenge in Multi-task learning, arising from task interactions that result in conflicting feature representations or optimization objectives [9]. Related tasks enable the network to find common and distinguishing points from tasks, thereby promoting deeper learning of the network. However, the issue of performance degradation in other tasks when optimizing one task in cross-task learning remains unresolved. This phenomenon often leads to inferior outcomes across tasks [10]. Therefore, a Multi-task classification framework is needed to achieve automatic EEG detection across tasks.

Knowledge distillation (KD) is a widely used technique in Multi-task learning that facilitates the transfer of knowledge from a teacher model to a lightweight student model. KD allows the student model to attain high performance with reduced computational resources [11]. In Multi-task learning, KD can alleviate task conflicts by allowing Single-task teacher models to guide the Multi-task student model. Traditional KD methods typically rely on pre-trained teacher models to provide supervisory signals, such as logits or intermediate representations, which guide the training procedure of the student model [12]. However, these methods face significant challenges that include increased training costs and difficulties in adapting to dynamic learning tasks [13]. Besides, Online knowledge distillation (OKD) has emerged as an innovative approach that eliminates the need for pre-trained teacher models by enabling the simultaneous training of teacher and student models [14]. OKD allows for real-time

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interaction between models, facilitating dynamic knowledge exchange throughout the training process [15]. In addition to enhancing task-specific performance, OKD significantly reduces inference time and memory requirements by jointly optimizing Multi-task and Single-task models, making it well-suited for resource-constrained environments. Therefore, we need a Multi-task classification framework with online feature distillation to achieve accurate automatic detection across tasks.

A. Limitation

As highlighted in the above analysis, it is evident that a wide range of researchers have explored Multi-task learning, distillation learning, and online distillation methods for various applications. However, the key issues associated with these methods are as follows:

- **Limited Task Utilization in Multi-Task Optimization:** In Multi-task learning, the optimization requirements of different tasks can vary significantly. Traditional static weighting strategies often fail to adapt dynamically to these requirements, leading to certain tasks dominating the optimization process while others are under-optimized.
- **Insufficiency of Cross Task EEG Knowledge Transfer:** The challenge in Cross Task EEG classification stems from the difficulty of effectively sharing and optimizing features across tasks. Cross Task learning frameworks inherently face tasks interference. Consequently, this leads to poor shared feature learning. Specifically, knowledge transfer between Single-task and Multi-task models often faces efficiency limitations is often limited, making it difficult to dynamically adapt to the diverse requirements of different tasks.
- **Poor Cross-Task Network Regulation in Multi-Task Learning:** In Multi-task learning, the optimization requirements of different tasks can vary significantly. Traditional static weighting strategies often fail to adapt dynamically to these requirements, leading to certain tasks dominating the optimization process while others are under-optimized.

B. Paper Contribution

In this paper, we proposed A novel Multi-task Learning-based Online Knowledge Distillation for Cross-task EEG Classification framework(MOKD-EEG). MOKD-EEG ensures robust feature sharing and supports adaptive task optimization. To achieve this, this paper propose three key modules: Online Knowledge Distillation Training(OKD-T), Adaptive Representation Distillation (ARD), Dynamic Weight Adjustment (DWA). Initially, OKD-T refines the distillation process by adaptively fine-tuning task-specific components based on task complexity and inter-task interactions. Furthermore, ARD dynamically adjusts intermediate feature representations between Single-task and Multi-task models. It reduces conflicts and enhances shared representation learning. Finally, DWA adjusts the contribution of each task during training by dynamically

weighting task-specific losses, ensuring balanced optimization across diverse EEG tasks. Therefore, MOKD-EEG not only provides an efficient and adaptive solution for cross-task EEG classification but also significantly advances the capabilities of BCI systems in diverse applications. The experimental outcomes show that our proposed method achieves impressive results. To summarize, this study highlights several important contributions:

- A novel Multi-task Learning-based Online Knowledge Distillation Training(OKD-T) Method was developed to address the challenges of cross-task variability in EEG classification.
- The ARD module dynamically adjusts feature distillation to reduce task interference and improve feature alignment.
- Task-specific prioritization during training is achieved through the DWA mechanism, ensuring balanced optimization across tasks.
- To validate the effectiveness of MOKD-EEG, experiments on Multiple EEG datasets demonstrated the framework's ability to improve cross-task classification performance and robustness.

The subsequent sections of this paper are structured as follows: Section II introduces the overall materials and methods. Datasets and experimental details are described in Section III. An exhaustive analysis of the overall experimental results is presented in Section IV. Section VI provides a conclusion.

II. MATERIALS AND METHODS

This section starts with an overview of our designed network. Next, we detail the proposed architecture. Finally, we elaborate on the carefully developed loss function.

Fig.1 illustrates the MOKD-EEG architecture, which includes three steps: Data Pre-processing, Online Knowledge Distillation Training, and Testing on Cross-task Dataset. Raw EEG signals are transformed into a standardized format suitable for training at first. The second procedure is that Online Knowledge Distillation Training serves as the core of the MOKD-EEG framework. Additionally, the method integrates a shared Vision Transformer (ViT) backbone into a Multi-task network that includes task-specific classification heads. The shared backbone extracts general spatial-temporal features, while the classification heads handle task-specific predictions. Furthermore, Adaptive Representation Distillation (ARD) is incorporated to dynamically align intermediate feature representations between Single-task and Multi-task models, enhancing shared feature learning. Dynamic Weight Adjustment (DWA) ensures balanced optimization by dynamically adjusting task-specific loss weights during training. The third part is testing the trained model on unseen cross-task EEG data to evaluate its performance. At last, the shared backbone and task-specific classification heads are used to process the test data and generate predictions for each task. This part validates the framework's ability to generalize across tasks. It showcases its robustness and adaptability in real-world applications.

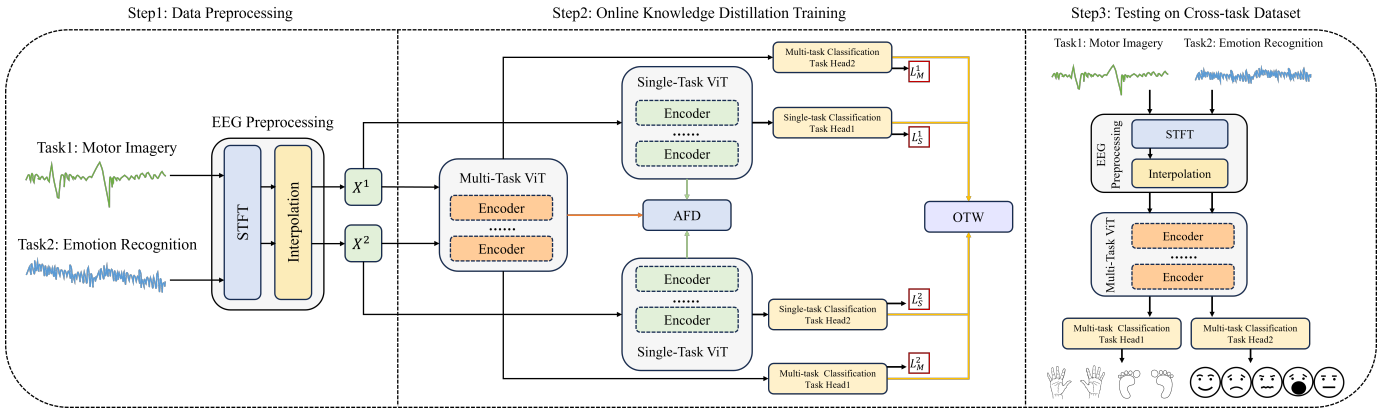


Fig. 1: The framework of MOKD-EEG.

Section II-A describes the Data Pre-processing stage. Section II-B introduces the Online Knowledge Distillation Training stage. Finally, Section II-D integrates contributions from both stages for end-to-end optimization.

A. Data Pre-processing

Transforming EEG signals into images through Short-Time Fourier Transform (STFT) enables the extraction of rich temporal and frequency-domain features, which are critical for improving neural network performance [16]. Additionally, EEG spectrograms allow for accurate visual representation, facilitating robust image-based classification. Therefore, using STFT to convert EEG into brain map representations enhances feature extraction and compatibility with image-based learning models, as shown in figure 2.

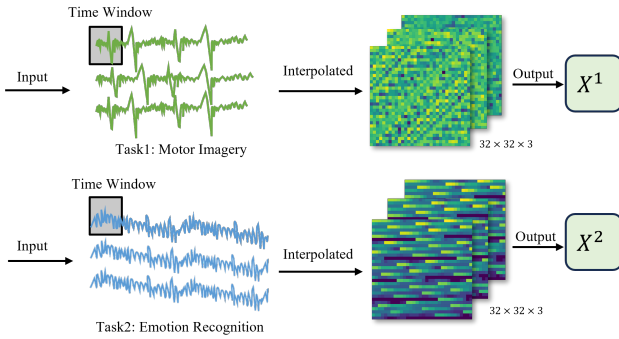


Fig. 2: The Process of Data Pre-processing.

Pre-processing raw EEG signals is a well-established method to improve the performance of Multi-task models. Consequently, data pre-processing plays a crucial role in advancing research within the field of brain-computer interfaces (BCI). Short-Time Fourier Transform (STFT) has significant advantages in EEG signal processing. It can analyze signals in the time-frequency domain, capturing dynamic changes in the spectrum while preserving important time and frequency features [17]. Besides, raw EEG signals from Motor Imagery and Emotion Recognition tasks are preprocessed using the STFT method to handle their non-stationary nature and improve their

compatibility and efficiency for subsequent tasks. The first step Pre-processing involves the following processes:

- **STFT:** Motor Imagery and Emotion Recognition task EEG raw data undergoes STFT with a frame length of 512 sample points, generating time-frequency representations. The results are log-scaled to enhance dynamic range and ensure non-negativity.
- **Interpolation:** The STFT outputs are reshaped into a 3D matrix (channels, freq_bins, time_frames) and linearly interpolated to standardize the frequency dimension to 1024 sample points. The final output consists of data for two tasks, denoted as X^1 and X^2 .

B. Online Knowledge Distillation Training(OKD-T)

Task interference is a significant challenge in Multi-task learning. This challenge can lead to decreased performance in individual tasks within the Multi-task framework. To tackle this problem, OKD-T is proposed to transfer task-specific knowledge extracted from Single-task models. This knowledge is transferred to a shared Multi-task model during the training phase.

As shown in the Step2 of Fig. 1, the framework utilizes the unique feature representations learned by Multi-task model and distills them iteratively into the Single-task model. OKD-T consists of a Multi-task and Single-task Network for feature extraction, ARD for aligning intermediate representations, and DWA for balancing task-specific contributions.

1) **Multi-task and Single-task Network Architecture:** Fig.1 illustrates the structure of OKD-T, which features a Multi-task network built upon a ViT backbone and task-specific classification heads. Additionally, the design incorporates Single-task networks, each employing a ViT backbone with dedicated classification heads. Both the Multi-task and Single-task classification heads adopt a simple linear structure, consisting of a fully connected layer that maps the feature embeddings to the output classes. Multi-task and Single-task Network Architecture based on EEG for cross-task learning in Motor Imagery and Emotion Recognition. The Single-task networks and the Multi-task network are trained concurrently on multiple tasks. Moreover, the Single-task networks assist in optimizing the

Multi-task network during training. Simultaneously, the Multi-task network shares intermediate features with the Single-task networks. Next, we will provide a detailed introduction to the data flow of the two networks separately.

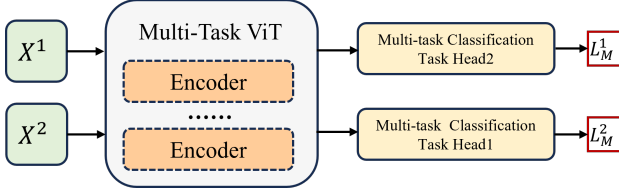


Fig. 3: The process of Multi-Task network.

Fig.3 illustrates the structure of the Multi-task ViT in OKD-T. Firstly, the final output of EEG Preprocessing module X^1 and X^2 are fed into the shared Multi-task network, which utilizes a ViT backbone to extract shared task features. Secondly, these shared features are passed to task-specific classification heads. The producing separate classification loss for Task A and Task B is represented as $\mathcal{L}_M^1$ and $\mathcal{L}_M^2$, where the number denotes the Task and M represents Multi-Task. We denote the feature representation at encoder layer l of the Multi-Task ViT as $\mathbf{f}_l^{\text{MTL}}$. *MTL* represents the learning of Multi-task networks. Finally, the classification loss for each task L_M^1 and L_M^2 are detailed:

$$\mathcal{L}_M^i = \mathcal{L}(\text{Linear}(\mathbf{f}_l^{\text{MTL}}(X^i, Y^i)), \mathbf{y}^i) \quad (1)$$

$$\mathcal{L}_M = \sum_{i=1}^2 \mathcal{L}_M^i \quad (2)$$

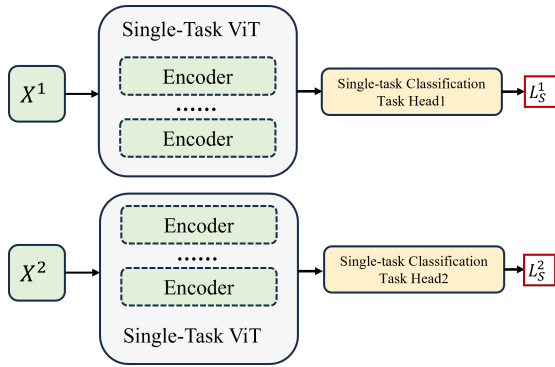


Fig. 4: The process of Single-task network.

In parallel, Task A and Task B data are also processed independently by their respective Single-task networks, as shown in Fig. 4. Each equipped with its own ViT backbone and classification head. We denote the feature representation at layer l of the Single-task ViT as $\mathbf{f}_l^{\text{STL}}$. *STL* represents the learning of Single-task networks. The classification loss for each task L_M^1 and L_M^2 is computed by comparing the classification head outputs with the corresponding ground truth labels as detailed:

$$\mathcal{L}_S = \sum_{i=1}^2 \mathcal{L}_S^i \quad (3)$$

$$\mathcal{L}_S^i = \mathcal{L}(\text{Linear}(\mathbf{f}_l^{\text{STL}}(X^i)), \mathbf{y}^i) \quad (4)$$

2) *Adaptive Representation Distillation (ARD)*: Multi-task learning faces a major challenge due to differences in feature distributions and task objectives. These methods often require distinct neural patterns to be captured, making it difficult for a shared model to generalize effectively across tasks [19]. Additionally, this challenge frequently leads to reduced performance and limited adaptability in Multi-task EEG classification systems. To address this, ARD is introduced as a mechanism that dynamically aligns intermediate feature representations between Single-task and Multi-task models, as the orange and green lines shown in Fig.1. ARD ensures that task-specific knowledge is preserved while optimizing shared feature representations, effectively mitigating the negative effects of task interference.

The feature representation at layer l of the MTL model is defined as $\mathbf{f}_l^{\text{MTL}}$, and the feature representation at layer l for the i -th STL model is defined as $\mathbf{f}_l^i$. To simplify processing, a new variable $\mathbf{d}_l$ is defined as the discrepancy between the MTL and weighted STL features:

$$\mathbf{d}_l = \mathbf{f}_l^{\text{MTL}} - \sum_{i=1}^{N_t} \omega_{l,i} \cdot \mathbf{f}_l^i, \quad (5)$$

where $\omega_{l,i}$ is a learnable parameter determining the contribution of task i 's features to the MTL representation at layer l . Building upon this, the ARD loss is expressed as:

$$\mathcal{L}_{\text{ARD}} = \sum_{l=1}^L \|\mathbf{d}_l\|^2. \quad (6)$$

During training, the weights $\omega_{l,i}$ are jointly optimized with the network parameters, dynamically adjusting the contribution of task-specific features to the shared representation. To prevent interference between Single-task and Multi-task training, gradients from $\mathcal{L}_{\text{ARD}}$ are detached from the computational graph of the STL networks. This ensures that the performance of STL models remains unaffected during the optimization process.

ARD employs adaptive weights to dynamically align intermediate feature representations between MTL and STL models. This layer-by-layer alignment ensures that shared representations are optimized to meet the unique requirements of each task, effectively addressing the negative transfer problem.

3) *Dynamic Weight Adjustment(DWA)*: In the yellow line of Fig.1, the proposed DWA is designed to dynamically adjust task weights during training. By comparing the effectiveness of the Multi-task model and Single-task models, DWA dynamically updates task weights during the training phase. It ensures that the MTL model prioritizes tasks where its performance lags behind the corresponding STL models, thereby improving overall training efficiency and reducing negative transfer.

At training iteration T , the task-specific losses for the MTL model and STL models are denoted as $L_i^{\text{MTL}}(t)$ and $L_i^{\text{STL}}(t)$, respectively. To quantify the relative performance difference between MTL and STL for the i -th task, we define a performance ratio m_i^t as:

$$m_i^t = \frac{L_i^{\text{MTL}}(t)}{L_i^{\text{STL}}(t)}. \quad (7)$$

The performance ratio is utilized to calculate the task weight $\lambda_i(t)$ for the i -th task using a temperature-scaled softmax function:

$$\lambda_i(t) = N_t \cdot \frac{\exp\left(\frac{m_i^t}{T}\right)}{\sum_{j=1}^{N_t} \exp\left(\frac{m_j^t}{T}\right)}. \quad (8)$$

where T is the temperature parameter that controls the sharpness of the weighting distribution, and N_t is the total number of tasks. A smaller T results in sharper weight distributions, emphasizing the tasks with larger performance gaps.

This dynamic weighting strategy ensures that tasks where the Multi-task model underperforms compared to STL models receive higher priority during training. The DWA mechanism effectively reduces the loss gap between Multi-task and Single-task models, as illustrated in experimental results, and aligns the optimization trajectory towards better Multi-task performance.

C. Testing on Cross-task Dataset

As illustrated in Fig.1, the final step involves testing the trained model on cross-task datasets to evaluate its generalization and adaptability. In this step, the preprocessed Cross-task EEG data are fed into the trained Multi-task ViT backbone. The backbone extracts shared spatial-temporal features, which are then passed to the task-specific classification heads for final predictions. Each classification head outputs the predicted labels corresponding to its respective task. This testing procedure demonstrates the model's effectiveness in generalizing across diverse tasks.

D. Loss Function

The overall loss function integrates the contributions of ARD and DWA to optimize the MOKD-EEG framework. The final loss is defined:

$$\mathcal{L} = \mathcal{L}_{\text{ARD}} + \sum_{i=1}^N (\mathcal{L}_i^{\text{STL}} + \lambda_i \mathcal{L}_i^{\text{MTL}}), \quad (9)$$

where:

- $\mathcal{L}_{\text{ARD}}$ is the Adaptive Feature Distillation loss, aligning feature representations between Single-task and Multi-task models.
- $\mathcal{L}_i^{\text{STL}}$ and $\mathcal{L}_i^{\text{MTL}}$ are the task-specific losses for Single-task and Multi-task models, respectively.
- λ_i is the task weight dynamically adjusted by the DWA module.

This integrated loss function ensures that the framework balances shared feature learning, task-specific performance, and inter-task interactions effectively.

III. EXPERIMENTAL SETUP

This section provides a detailed set of experiments to showcase the effectiveness of the proposed CD³Net. The description of the Cross-task Dataset is outlined in Section III-A, while the parameter configurations are discussed in Section III-B.

A. Dataset

1) *BCI-IV-2A*: The BCI Competition IV dataset 2A [20] comprises EEG data collected from nine participants performing four Motor Imagery (MI) tasks, including movements of the left hand, right hand, feet, and tongue. The recordings were obtained using 22 channels at a sampling rate of 250 Hz and were bandpass-filtered within the range of 0.5 to 100 Hz.

2) *SEED-V*: The SEED-V [21] is a video-induced emotion task EEG data set. The experiment involved 16 participants who were each exposed to 15 different stimuli on three separate occasions to elicit emotional responses. The 15 types of stimuli were designed to elicit five categories of emotions: disgust, fear, sadness, neutrality, and happiness. Physiological signal data were recorded from 62 electrodes using the 10-20 system, with a sampling rate set at 1000 Hz.

B. Parameter Settings

We use the ViT-Tiny [18] model as the backbone network. The parameters used in the loss functions are configured to ensure effective training and fair evaluation across tasks. For the performance ratio m_i^t defined in Equation 7, the task-specific losses $L_i^{\text{MTL}}(t)$ and $L_i^{\text{STL}}(t)$ are computed at each iteration t using the respective task-specific loss functions. The temperature parameter T in Equation 8 is set to 0.1, which provides a balance between focusing on tasks with larger performance gaps and maintaining stable optimization.

The ARD mechanism uses learnable weights $\omega_{l,i}$ for each layer l and task i , which are initialized to 1 and optimized jointly with the network parameters. The discrepancy d_l in Equation 5 is computed for all layers in the shared backbone, with L set to the total number of transformer layers, which is 12 for the ViT architecture used in our experiments.

For the overall loss function in Equation 9, the task-specific losses $\mathcal{L}_i^{\text{STL}}$ and $\mathcal{L}_i^{\text{MTL}}$ are computed using cross-entropy loss for semantic segmentation, L_1 loss for depth estimation, and cosine similarity loss for surface normal estimation. The weights λ_i for task-specific losses are dynamically adjusted at each iteration using the DWA mechanism.

This paper adopts ten fold cross-validation for all experiments. All models trained with the AdamW optimizer [22]. Based on multiple experimental results, the paper selects the following hyperparameters, as shown in Table I.

TABLE I: Datasets Hyperparameters List in Classification Model.

Dataset	BCI-IV-2A [20]	SEED-V [21]
Learning Rate	0.009	0.006
Batchsize	32	128
Dropout	0.01	0.1
Epoch	1000	1000

IV. RESULT

To evaluate the effectiveness of our approach, we perform a thorough assessment using Cross-task public datasets. First, we compare our model against various methods across multiple datasets. Then, we carry out ablation studies to validate the impact of our proposed contributions. Finally, we summarize the experimental results obtained from an extensive set of qualitative analyses.

SectionIV-A and SectionIV-B presents a comparative experiment using subject-independent methods on the Motor Imagery and Emotion Recognition task datasets. The results for the BCI-IV-2A and SEED-V datasets are shown in Tables II and III.

A. Performance on the Motor Imagery Task classification

TABLE II: Model performance BCI-IV-2A for subject-independent average accuracy.

Type	Method	Year	Avg ACC (%)
Transformer	Transformer-based [23]	2023	81.04
	SNMF [24]	2024	73.78
Hybrid Network	SSDA [25]	2024	61.00
	DB-ATCNet [26]	2024	71.30
	MultiscaleMCA [27]	2024	74.33
	MOKD-EEG	2025	81.74 $\pm$ 1.4

Table II shows that the listed methods can be generally divided into Transformer-based approaches and hybrid networks. The proposed MOKD-EEG model achieved the highest average accuracy of 81.74% $\pm$ 1.4%, outperforming both Transformer-based and hybrid network models. Additionally, it achieved a notable enhancement of 7.96% and 20.74% in accuracy when compared to the hybrid network models. These experimental findings illustrate the superior performance of MOKD-EEG, establishing it as an effective framework for subject-independent EEG classification in BCI-IV-2A tasks.

B. Performance on Emotion Recognition classification task

TABLE III: Model performance on SEED-V dataset for subject-dependent and average accuracy.

Type	Method	Year	Avg ACC (%)
TL	CD-EmotionNet [28]	2023	82.67
Hybrid Network	RHPRNet [29]	2024	55.84
	PGCN [30]	2023	82.24
	G2G+ResNet50 [31]	2023	83.87
	MOKD-EEG	2024	85.16 $\pm$ 7.51

From Table III, the approaches for subject-independent classification on the SEED-V dataset can be broadly categorized into transfer learning (TL) methods and hybrid networks. The proposed MOKD-EEG model significantly outperformed all competing methods, achieving an average accuracy of 90.16% $\pm$ 7.51%. These experimental results validate the superiority of MOKD-EEG over existing methods.

C. Ablation Studies

The proposed MOKD-EEG framework integrates three critical components: OKD-T, ARD, DWA. To validate the contributions of each component, ablation studies were conducted on the motor imagery and emotion recognition tasks. Initially, we evaluated a basic Multi-task learning model without OKD-T, ARD, or DWA. Next, OKD-T was introduced into the framework. The addition of OKD-T demonstrated the effectiveness of non-online knowledge distillation method in improving overall performance by enabling efficient knowledge transfer across tasks. Subsequently, we assessed the role of ARD by incorporating it into the model without DWA. This configuration ensured that intermediate features were aligned between Single-task and Multi-task networks, reducing task interference and enhancing shared feature representation. The results confirmed the critical role of ARD in mitigating conflicts across tasks. The specific results are described in detail below.

When the OKD-T module was disabled, the accuracy on the motor imagery task dropped from 81.74% $\pm$ 1.4% to 76.50% $\pm$ 2.01%. Similarly, on the emotion recognition task, the accuracy decreased from 85.16% $\pm$ 7.51% to 84.00% $\pm$ 3.7%. These results highlight the importance of OKD-T in effectively transferring knowledge across tasks.

When the ARD mechanism was removed, the accuracy on the motor imagery task dropped from 81.74% $\pm$ 1.4% to 78.70% $\pm$ 1.52% with a reduction of 3.04%. For the emotion recognition task, the accuracy decreased from 85.16% $\pm$ 7.51% to 83.60% $\pm$ 4.33%. These findings emphasize the role of ARD in aligning feature representations and mitigating task interference.

Excluding the DWA module resulted in a drop in accuracy on the motor imagery task from 81.74% $\pm$ 1.4% to 79.20% $\pm$ 1.37%. On the emotion recognition task, the accuracy decreased from 85.16% $\pm$ 7.51% to 84.60% $\pm$ 2.13%. This demonstrates the necessity of DWA for ensuring balanced task optimization.

When OKD-T and ARD were combined without DWA, the accuracy on the motor imagery task reached 80.00% $\pm$ 3.27%, showing a slight reduction of 1.74% compared to the full model. For the emotion recognition task, the accuracy was 85.40% $\pm$ 8.19%, a decrease of 0.24%. These results suggest that OKD-T and ARD play complementary roles, even without the inclusion of DWA.

The full MOKD-EEG framework, integrating OKD-T, ARD, and DWA, achieved the best performance on both tasks, with accuracies of 81.74% $\pm$ 1.4% for motor imagery and 85.16% $\pm$ 7.51% for emotion recognition. These results

TABLE IV: Ablation Study on MOKD-EEG Framework.

Method			Performance			
OKD-T	ARD	DWA	Motor Imagery Task (%)	Improvement (%)↑	Emotion Recognition (%)	Improvement (%)↑
×	×	×	74.50 ± 1.8	-	82.00 ± 6.5	-
✓	×	×	76.50 ± 2.01	2.00	84.00 ± 3.7	2.00
×	✓	×	78.70 ± 1.52	4.20	83.60 ± 4.33	1.60
×	✓	×	80.33 ± 3.27	5.50	85.40 ± 8.19	3.40
✓	×	✓	79.20 ± 1.37	4.70	84.60 ± 2.13	3.20
✓	✓	×	81.47 ± 3.5	5.50	84.50 ± 7.78	3.50
✓	✓	✓	81.74 ± 1.4	7.24	85.16 ± 7.51	8.16

validate the combined effectiveness of all components in addressing cross-task EEG classification challenges.

V. DISCUSSION AND FUTURE WORKS

This paper proposed the MOKD-EEG framework to address the challenges of cross-task EEG classification. The framework integrates preprocessing with STFT and a ViT-based backbone for shared feature extraction. Additional mechanisms such as ARD and DWA are employed to enhance task performance. Experiments on multiple public datasets demonstrated that the proposed approach achieved superior performance across motor imagery and emotion recognition tasks compared to existing methods. Ablation experiments demonstrated the value of each individual component and highlighted the impact of dynamic knowledge distillation and task prioritization.

However, the fixed STFT window size in preprocessing limits the model's ability to capture rapid signal changes, and the focus on unimodal EEG data reduces the richness of feature representations. Lastly, while ARD and DWA effectively mitigate task interference and balance priorities, exploring more sophisticated strategies for feature alignment and task optimization could further enhance performance.

Based on what have mentioned above, future work will aim to address these limitations by improving preprocessing techniques, such as exploring adaptive time-frequency transformation methods. Furthermore, incorporating multimodal data, such as EEG and eye movement, could enrich feature representations and enhance classification robustness. Finally, extending the framework to include advanced task-adaptive mechanisms may further optimize performance in dynamic and complex EEG applications.

VI. CONCLUSION

This work introduces the MOKD-EEG framework, a novel architecture designed to address the challenges of cross-task EEG classification by leveraging flexibility and adaptability in Multi-task learning. The framework effectively utilizes pre-processed EEG data through STFT to capture comprehensive temporal and spectral features. Furthermore, OKD-T based on the ViT-based backbone facilitates the extraction of shared spatial-temporal features, enhancing the generalization across tasks. The integration of ARD ensures efficient alignment of task-specific and shared feature representations, while DWA dynamically balances task prioritization during training.

The proposed MOKD-EEG framework has been comprehensively evaluated on cross-task public datasets, showcasing its ability to generalize effectively across diverse EEG tasks, such as motor imagery and emotion recognition. Results demonstrate that the identification of essential EEG patterns relies heavily on optimized feature representations and task-specific adaptations. MOKD-EEG outperforms existing methods, setting a new benchmark for cross-task EEG classification, and providing a robust solution for brain-computer interface applications.

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